The Fractional Distribution and Tilted Stable Law

From the Chinese Restaurant to Synthetic Regime Data

Stephen Lihn (2026)

Source: *Fractional Distributions* (fracdist)

Reference implementation: `github.com/slihn/(gas-impl, Mixture-VAE)`



v113

The challenge in financial time series

The time series is often not long enough for machine learning models.
We need to use **synthetic** data for pre-training.

If we choose the **parametric approach**, we need a good parametric distribution.
Financial assets often show large kurtosis and sharp peak density.
It is hard to find the distribution that describes the **shape and moments** properly.

Issues in the existing data generator for regime detection:

- Gaussian emissions make the problem **too easy** — every model wins.
- Student-*t* gives fat tails, but only **one** parameter (df) to tune.
- When tails are **too fat**, we need to clip the tails (clip_factor).

Main results

Generalized α-stable distribution (GAS)

- It **fits financial return data** well. Clipping is (almost) unnecessary.
- It can take on **any ticker** with 10-20 years of history!
- Its symmetric distribution (GSAS) has rich structures, akin to **Central Limit Theorem**
- It has an elegant connection to the **tilted stable law** (Pitman-Yor, Devroye, James)

Benchmarked the two-state result on Yuqi's regime-detection GitHub project

- Mixture-VAE, Jump, KMeans++
- Achieved high accuracy in separating the bull/bear states

GAS could be THE distribution family for asset returns

- This is a big claim !!! Will be explained in the next few slides with a focus on **GSAS**.
- Its parameters have meanings in classic statistical distribution.

GSAS: symmetric, two shape parameters (α, k)

- GSAS is a normal mixture: $\sim N/V$ where V is the fractional chi (FCM) $\bar{\chi}_{\alpha,k}$.
- α: stability index $(0, 2]$, k: degrees of freedom $\in \mathbb{R}_{\neq 0}$, $\zeta = \text{sign}(k)$
- V is a squared tilted stable law T: $V \overset{d}{=} \sigma^{\zeta} T_{\alpha/2, z/2}^{-\zeta/2}$
  - $z = k - 1$ when $k \geq 1$: generalizes α-stable, Student's t (original branch)
  - $z = |k|$ when $k \leq 0$: generalizes exponential power (characteristic branch)

GAS-SN: GSAS + skew

- add β from skew-normal $N \sim SN(\beta)$, generalizes skew-t

GSAS version of CLT

The canonical coordinates

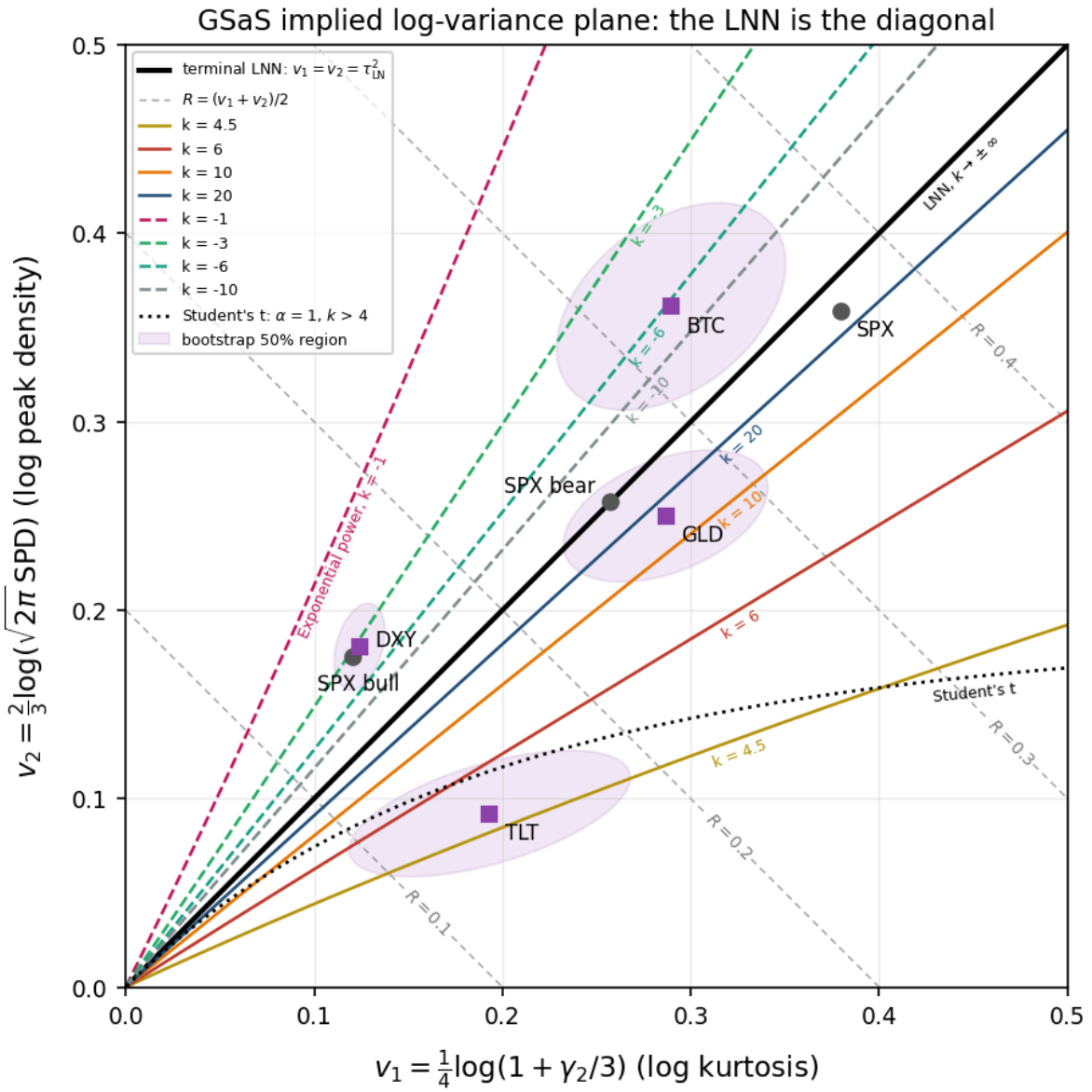
X-axis: excess kurtosis

Y-axis: std peak density

- LNN limit $(k \to \pm\infty)$ at diagonal (thick black)
- Normal at origin
- Fits SPX, bull, bear
- Fits GLD, DXY, TLT, BTC

Challenge

- In the small α, large k region
- Numerically harder



Fitting financial data becomes easy

- GLD, DXY, TLT, BTC
- Step 1: $(v_1, v_2) \to (\alpha, k)$
- Step 2: fits mean, sd

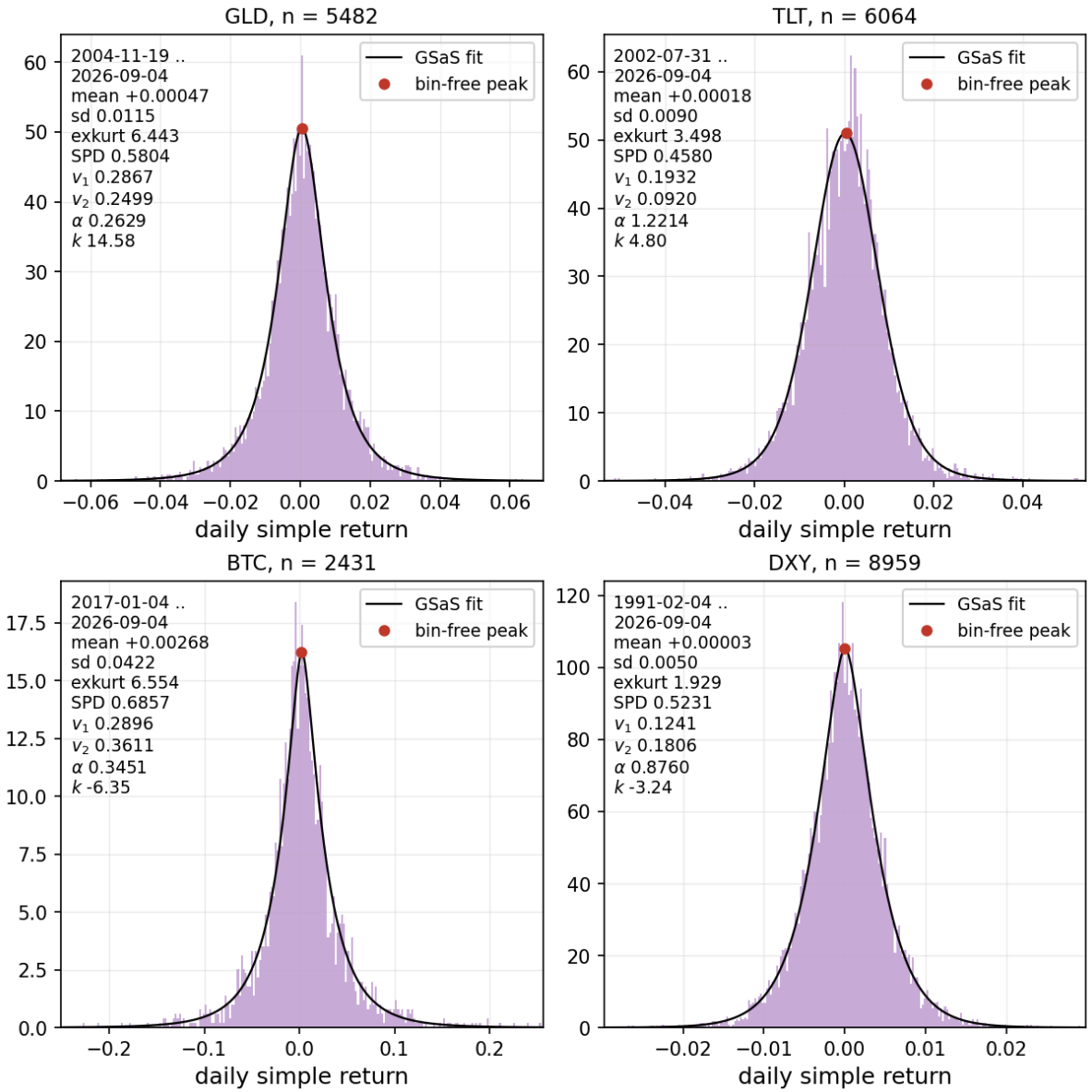
Then you can

- add skewness (GAS-SN)
- apply MLE

Why does it work so well?

A generalized method of moments:

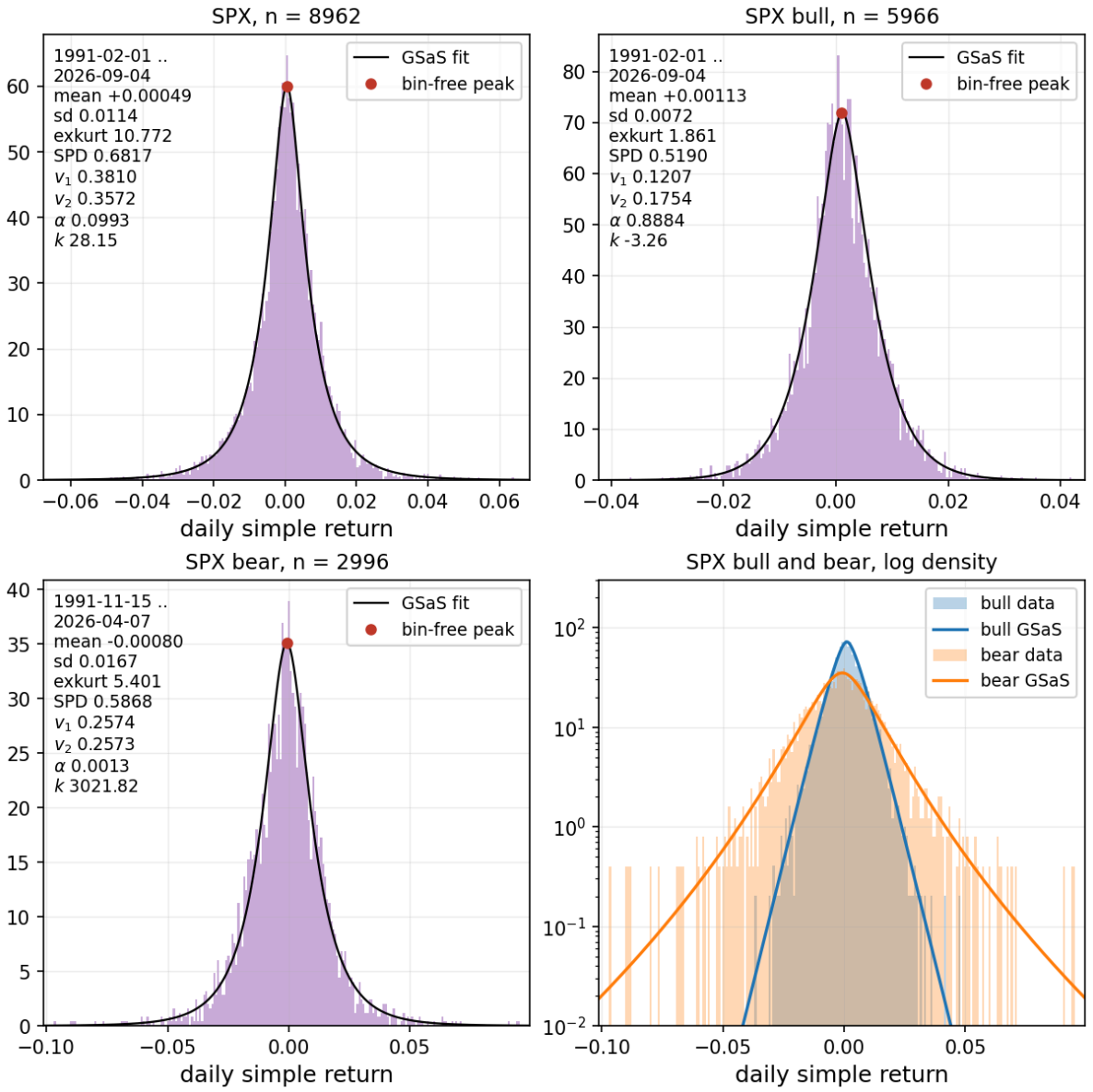
It fits $\mathbb{E}[V^n]$ where $n = 1, -2, -4$
($\mathbb{E}[V^0]$ is a constant. Only $-1, -3$ are left out.)



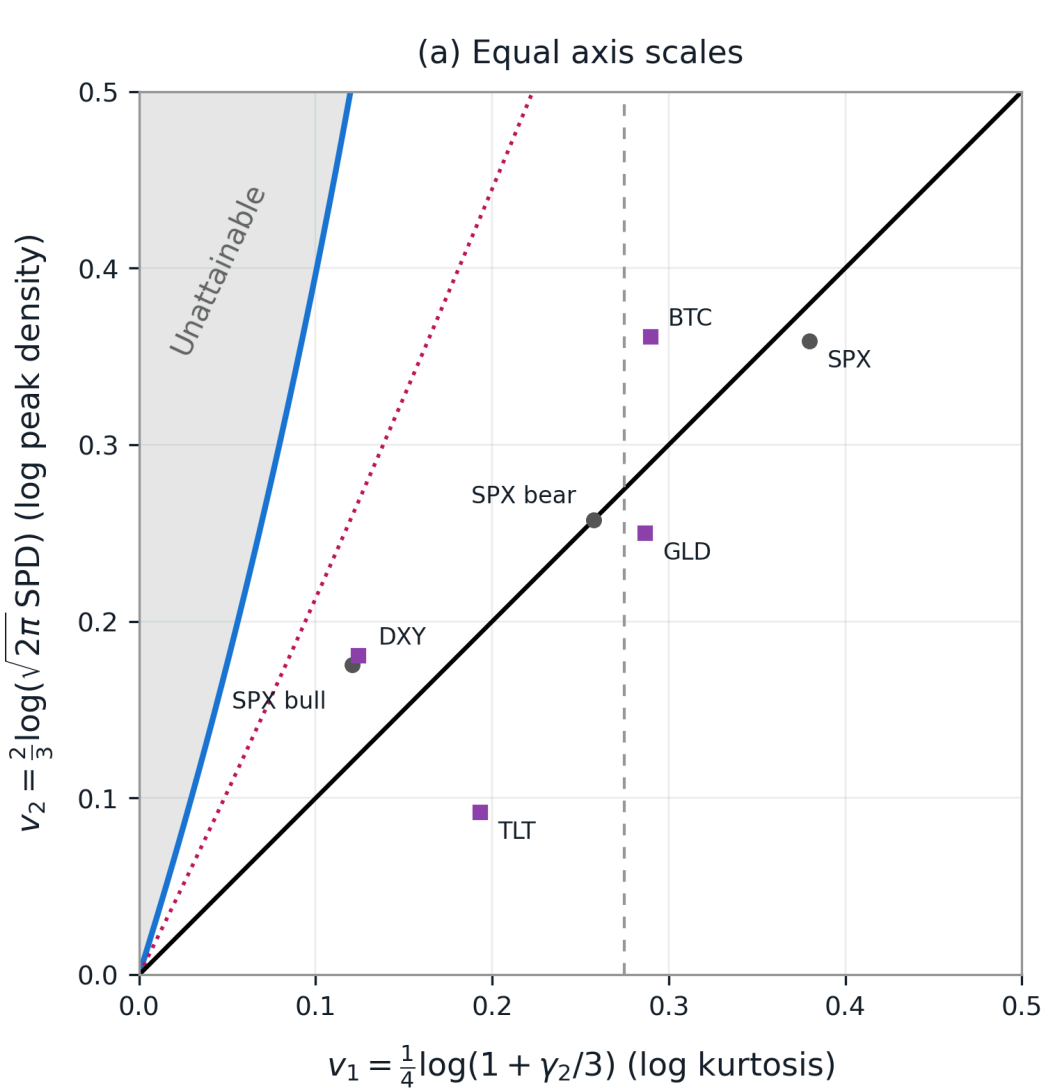
Fitting financial data becomes easy

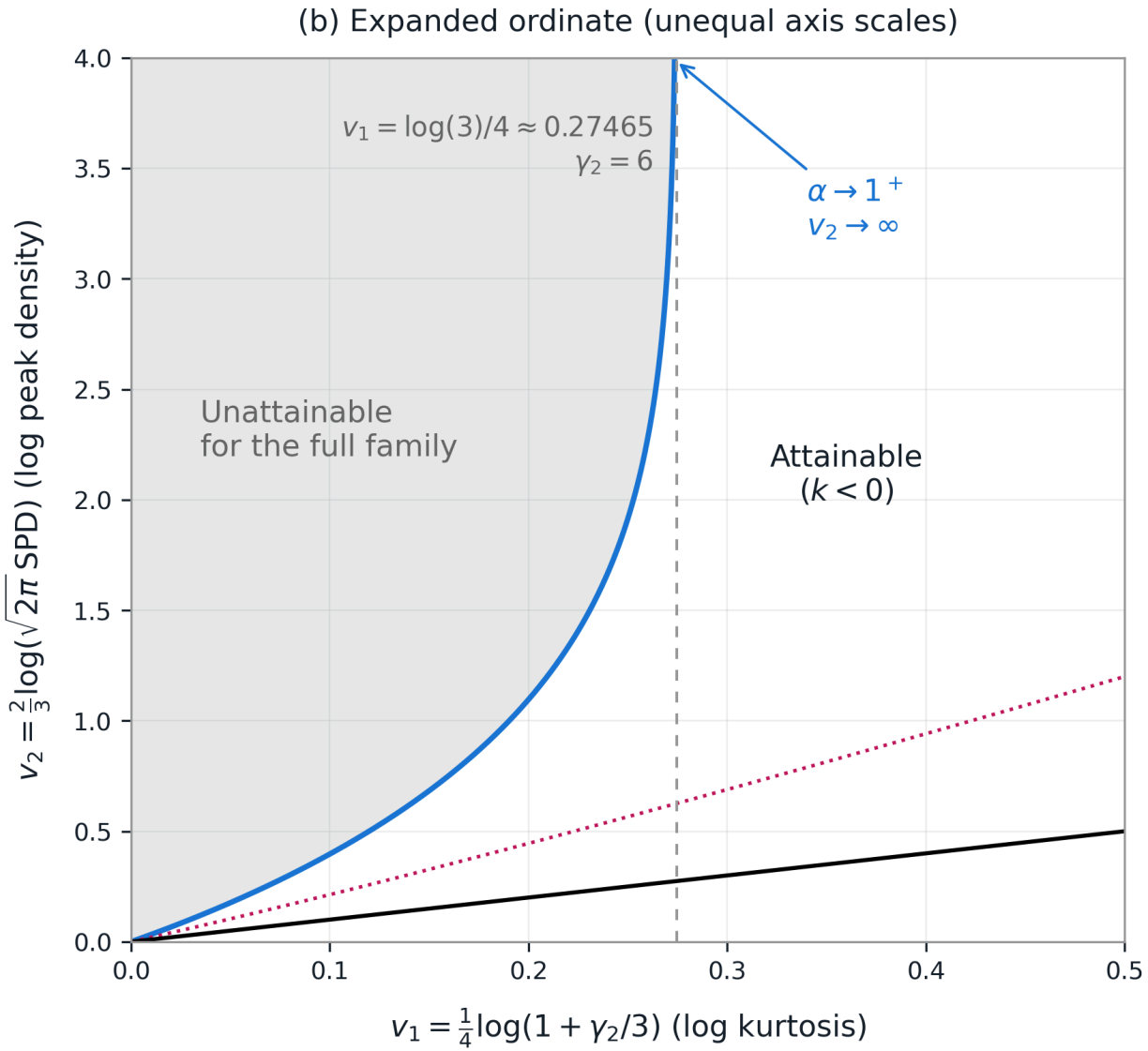
- SPX, bull, bear
- Two regimes lower the kurtoses
- Bull at $k \approx -3$: thin tails
- Bear near/on the LNN terminal line

Note: Study of the bear state led to the LNN limit in 9/2026.



GSAS has a very large admissible space





The lognormal-normal (LNN) limit

Roughly speaking, let $k \to \pm\infty$ and $\alpha \to 0$ with $1/(\alpha|k|) \to \tau_{\mathrm{LN}}^2$ fixed.
Then the FCM $V \sim \bar{\chi}_{\alpha,k}$ has a lognormal limit:

$$\frac{V}{e^{\mathbb{E}[\log V]}} \xrightarrow{d} \mathrm{LogNormal}(0, \tau_{\mathrm{LN}}^2).$$

So GSAS $X = N/V$ has an LNN limit after normalization:

$$X\, e^{\mathbb{E}[\log V]} = \frac{N}{V/e^{\mathbb{E}[\log V]}} \xrightarrow{d} N\, e^{\tau_{\mathrm{LN}} W} = \mathrm{LNN}(\tau_{\mathrm{LN}}^2),$$

with N, W independent standard normals.

A natural coordinate system: the LNN is the diagonal, with $k > 0$ below and $k < 0$ above it.
The **inverse degrees of freedom** $\xi \approx 1/k$ is the signed distance from it: $\xi = 0$ is the LNN.

FCM's lognormal limit is a form of "tempered stable"

Tempered stable law is frequently used in financial models.
It adds a $e^{-\lambda_{TS} r}$ factor to the Lévy measure to **temper** the fat tails.

Near the lognormal limit ($\xi \approx 1/k \to 0$), the centered log FCM, $U = \log V - \mathbb{E}[\log V]$, is additively infinitely divisible and has the approximate cumulants $(n \geq 2)$,

$$\kappa_n = (n-2)!\, \tau_{LN}^2 \, (-\xi)^{-(n-2)} \left[1 + O(|\xi|^3)\right],$$

which is a centered **tempered stable law**, with jump-magnitude Lévy measure

$$\nu_U(dr) = \delta \frac{e^{-\lambda_{TS} r}}{r^{1+\alpha_{TS}}}\, dr, \quad r > 0$$

where $\alpha_{TS} = 1$, $\delta = \tau_{LN}^2/|\xi| \gg 1$ and $\lambda_{TS} = |\xi|^{-1} \gg 1$. Jump direction: $-\xi$.

This is the so-called "Classic Tempered Stable" (CTS) distribution at index one.

(Rosinski 2007, Massing 2024, R package TempStable)

The Combinatoric Origin of The Tilted Stable Law

The two-parameter tilted stable law

It arose from Pitman & Yor's study of the **Poisson–Dirichlet** distribution.

$$T_{\alpha,\beta}^{-\alpha} = (T_{\alpha,\beta})^{-\alpha}$$

- $\alpha \in [0,1]$ — the **stability index**
- $\beta \geq 0$ — the power of the **polynomial tilt**
- $L_\alpha := T_{\alpha,0}$ — the one-sided stable law, often characterized by the Laplace transform:

$$\mathbb{E}\left[e^{-tL_\alpha}\right] = e^{-t^\alpha}, \qquad t \geq 0.$$

The **polynomial tilt** β is added to the density $L_\alpha(x)$ of L_α as

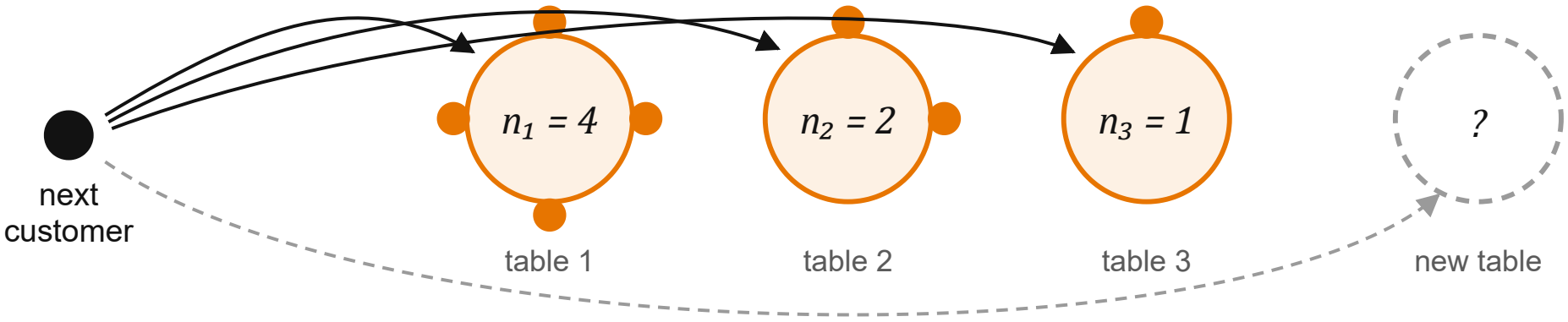
$$f_{T_{\alpha,\beta}}(x) \;\propto\; x^{-\beta} L_\alpha(x), \qquad x > 0.$$

Where it comes from - the single-parameter model

The Chinese restaurant process (CRP), one customer at a time

Given $\alpha \in [0, 1]$, K_n tables occupied after n customers; n_j customers already at table j:

- Choose a large table to join? $\Pr(\text{join table } j) = \frac{n_j - \alpha}{n} \longrightarrow$ Preferential attachment.
- Open a new table? $\Pr(\text{new table} \mid K_n) = \alpha \frac{K_n}{n} \longrightarrow$ Diversity, creativity (or anti-social).
- The α-diversity limit is $\frac{K_n}{n^\alpha} \longrightarrow M_\alpha := L_\alpha^{-\alpha}$ (the inverse stable law)



The two-parameter Pitman-Yor process in CRP representation

K_n tables occupied after n customers; n_j customers already at table j:

$$\Pr(\text{join table } j) = \frac{n_j - \alpha}{\beta + n}, \qquad \Pr(\text{new table} \mid K_n) = \frac{\beta + \alpha K_n}{\beta + n}$$

- α plays a **discount** role — larger α discounts the pull of existing tables.
- β is a **baseline strength** for creating a new table, independent of n and K_n.

The **α-diversity limit** is:

$$\frac{K_n}{n^\alpha} \longrightarrow T_{\alpha,\beta}^{-\alpha}$$

(Combinatorial Stochastic Processes, Pitman, 2002)

The three-parameter tilted stable law: $\mathrm{TS}(\alpha, \beta, \gamma)$

The superscript (exponent) is generalized to $T_{\alpha,\beta}^{-\gamma}$ with $\gamma \in \mathbb{R}$ in Devroye (2009):

$$\mathrm{TS}(\alpha, \beta, \gamma) \;:=\; T_{\alpha,\beta}^{-\gamma} = (T_{\alpha,\beta})^{-\gamma}$$

- $\gamma > 0$ — represents a **local time** process
- $\gamma < 0$ — represents a **volatility** process

The **bridge** to the fractional distribution -

> *"This three-parameter distribution law is **identical to our fractional gamma distribution** with a different parameterization."*

One foundational element, eight one-sided distributions

$T_{\alpha,\beta}^{-\gamma}$ is the foundational element that builds all one-sided distributions.

Fractional distribution	§	Notation	α	β	γ	scale
One-sided stable	4.2	L_α (aka $T_{\alpha,0}$)	α	0	-1	1
M-Wright / Mittag-Leffler	3.4	M_α	α	0	α	1
Fractional gamma	6.5	$N_\alpha(\sigma, d, p)$	α	$\alpha d/p$	α/p	σ
Fractional chi (FCM)	7.2.1	$\bar{\chi}_{\alpha,k}$, $k > 0$	$\alpha/2$	$(k-1)/2$	$1/2$	$\sigma_{\alpha,k}$
Inverse FCM	9.4	$\bar{\chi}_{\alpha,k}^{\dagger}$	$\alpha/2$	$(k-1)/2$	$-1/2$	$\sigma_{\alpha,k}^{-1}$
Characteristic FCM	7.4.2	$\bar{\chi}_{\alpha,-k}$	$\alpha/2$	$k/2$	$-1/2$	$\sigma_{\alpha,k}^{-1}$
FCM2	7.6	$\bar{\chi}_{\alpha,k}^{2}$	$\alpha/2$	$(k-1)/2$	1	$\sigma_{\alpha,k}^{2}$
Characteristic FCM2	7.6	$\bar{\chi}_{\alpha,-k}^{2}$	$\alpha/2$	$k/2$	-1	$\sigma_{\alpha,k}^{-2}$

- In the last line: $\bar{\chi}_{\alpha,-k}^{2}$ is $T_{\alpha/2,k/2}$. The reparametrization is most obvious here.

FG: The fractional gamma — Overview

$$N_\alpha(x;\sigma,d,p) \;:=\; C\left(\frac{x}{\sigma}\right)^{d-1} F_\alpha\!\left(\left(\frac{x}{\sigma}\right)^{p}\right), \qquad x \geq 0$$

where $F_\alpha(x) := W_{-\alpha,0}(-x) = \alpha x M_\alpha(x)$ is the Wright function of the second kind.

- $\alpha \in [0,1]$ — shape of the Wright function
- σ — scale; p — tail shape ($p \neq 0$, $dp \geq 0$); d — degrees of freedom

FG is the fractional extension of the generalized gamma $\mathrm{GG}(a,d,p)$: replacing $e^{-(x/a)^p}$ by the Wright function $F_\alpha((x/\sigma)^p)$. GG is recovered at $\alpha = 0$, and — more usefully — at $\alpha = \frac{1}{2}$, which is the line that "leads to the fractional extension of the χ distribution."

$$C = \frac{|p|}{\sigma}\,\frac{\Gamma(\alpha d/p)}{\Gamma(d/p)} \qquad (\alpha \neq 0,\; d \neq 0)$$

Fractional gamma *is* the tilted stable law (7/2026)

The FG random variable is exactly the re-parametrized tilted stable variable.

Let $X \sim N_\alpha(\sigma, d, p)$. Then

$$\boxed{X = \sigma\, T^{-\alpha/p}_{\alpha,\, \alpha d/p}}$$

i.e. $\beta = \alpha d/p$ and $\gamma = \alpha/p$.

- The polynomial tilt β is the "degree of freedom" parameter in FG and χ.

This mapping is an important finding. Everything downstream — FCM, FCM2, fractional F, GSAS, GAS-SN — inherits both its **analytics** (Mellin transform, moments) and its **sampler** from this one line.

Sampling by modified Kanter's method

$$T_{\alpha,\beta} = A_\alpha(Q_\beta)\, G_\beta^{-c}, \qquad c = \frac{1-\alpha}{\alpha}$$

1. $A_\alpha(q)$ — the **Zolotarev function**
2. Q_β — the tilted angle, with density on $(0, 1)$

$$f_{Q_\beta}(q) = I_\beta A_\alpha(q)^{-\beta}, \qquad I_\beta = \frac{\Gamma(1+\beta)\Gamma(1+c\beta)}{\Gamma(1+\beta/\alpha)}$$

3. $G_\beta \sim \mathrm{Gamma}(1 + c\beta,\ \mathrm{scale} = 1)$

Then the tilted stable variable U is

$$U = T_{\alpha,\beta}^{-\alpha} = A_\alpha(Q_\beta)^{-\alpha}\, G_\beta^{1-\alpha}.$$

The FG variable is $X = \sigma\, U^{1/p}$. The FCM variable follows.

Two draws and a closed-form transform. No numerical CDF inversion. Fast.

FCM: The fractional χ-mean

FCM is defined in two branches of k, so that the negative branch is the characteristic transform of the positive branch:

- Positive branch: $\bar{\chi}_{\alpha,k}(x) := N_{\alpha/2}\big(x;\ \sigma = \sigma_{\alpha,k},\ d = k-1,\ p = \alpha\big)$;
- Negative branch: $\bar{\chi}_{\alpha,-k}(x) := N_{\alpha/2}\big(x;\ \sigma = \sigma_{\alpha,k}^{-1},\ d = -k,\ p = -\alpha\big)$,
  where $\sigma_{\alpha,k} = \left(|k|^{1/2-1/\alpha}\right)/\sqrt{2}$ is the mean scaling factor and $\alpha \in (0,2)$ here.

The two branches are combined into a single expression for $k \in \mathbb{R}(k \neq 0)$:

$$\boxed{\bar{\chi}_{\alpha,k} = \sigma^{\zeta} T^{-\zeta/2}_{\alpha/2,z/2}}$$

where $\zeta = \mathrm{sign}(k)$, and $z = k-1$ for $k \geq 1$, $z = |k|$ for $k < 0$.

The second element: Selective sampling

One-sided elements give tails. **Skewness needs a second mechanism** (§1.2, §14.4).

Take $X_0 \sim N_d(0, \bar{\Omega})$, $X_1 \sim N(0,1)$, skew parameter $\beta \in \mathbb{R}^d$.
The skew-normal (SN) variable is (Azzalini (2013))

$$Z = \begin{cases} X_0 & \text{if } X_1 > \beta^\intercal X_0 \\ -X_0 & \text{otherwise} \end{cases} \qquad \Longrightarrow \qquad Z \sim SN_d(0, \bar{\Omega}, \beta)$$

A two-sided distribution is assembled as a **ratio** $Z/\mathrm{TS}(\cdot)$ or a **product** $Z \times \mathrm{TS}(\cdot)$.

The **crown jewel** is the **skew multivariate elliptical** distribution, based on the ratio

$$SN_d(0, \bar{\Omega}, \beta)/\bar{\chi}_{\alpha,k}.$$

GAS-SN: The univariate two-sided skew distribution (Ch. 12)

Definition 12.1: Let $Z \sim SN(0, 1, \beta)$ and $V \sim \bar{\chi}_{\alpha,k}$. Then

$$\boxed{X = Z/V \ \sim \ L_{\alpha,k}(\beta)\ , \quad \alpha \in (0, 2], k \in \mathbb{R}, \beta \in \mathbb{R}}$$

$$\text{PDF:} \quad L_{\alpha,k}(x; \beta) = 2 \int_0^\infty \mathcal{N}(xs)\, \Phi_{\mathcal{N}}(\beta x s)\, \bar{\chi}_{\alpha,k}(s)\, s\, ds$$

It is a **continuous mixture** — one skew-normal per value of the mixing variable V.

Limit	Reduces to
$k = 1, \beta = 0$	Paul Lévy's original symmetric α-stable dist
$\beta = 0$	The generalized symmetric α-stable dist, called "GSAS", $L_{\alpha,k}$
$\alpha = 1, k > 0$	Azzalini's skew-***t***: $T(\beta, k) = L_{1,k}(\beta)$
$1/(\alpha\lvert k\rvert)$ fixed and $\lvert k\rvert \to \infty, \beta = 0$	the lognormal-normal distribution (LNN)
$\alpha \to 2$ or $\lvert k\rvert \to \infty, \beta = 0$	the normal distribution $\mathcal{N}(0, 1)$

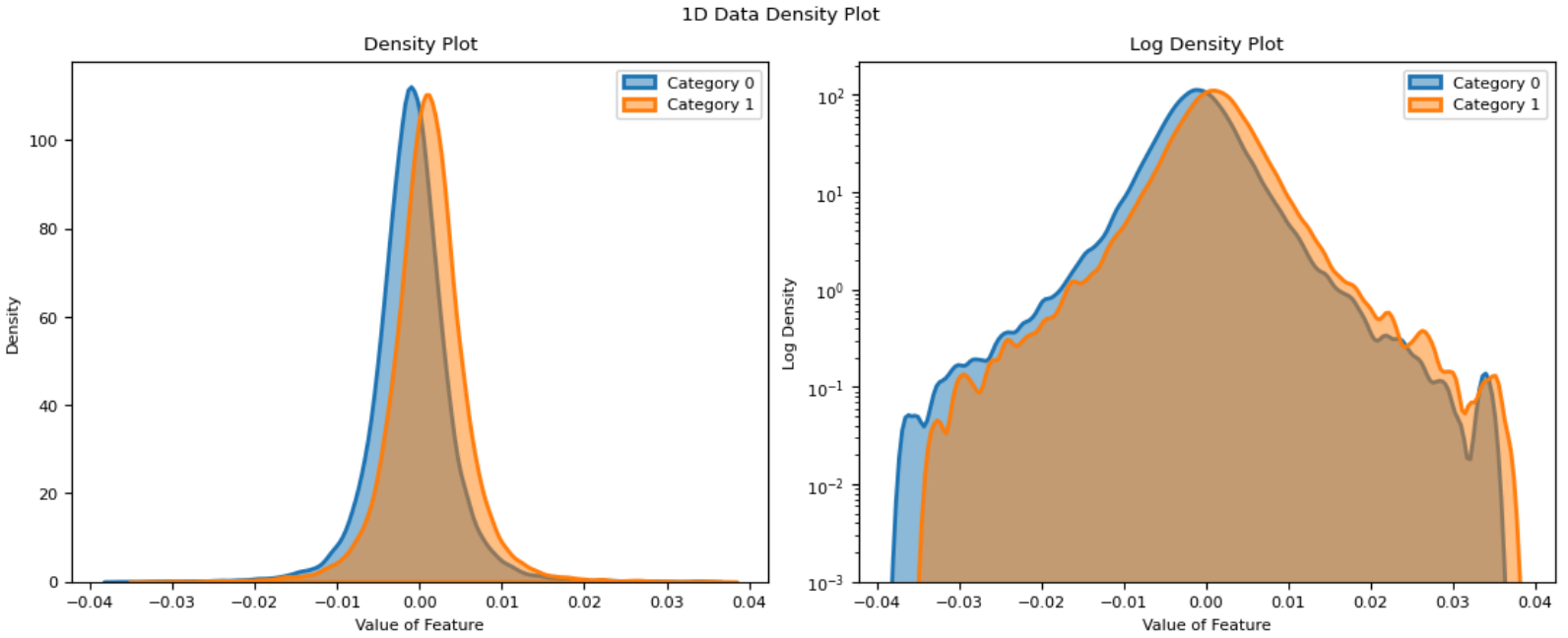
Synthetic Time Series and Regime Detection

From variates to synthetic regime data

Yuqi's Mixture-VAE code base: (Nie, Mulvey, Poor, Yu & Huang (2026))

```
data_code/synthetic_data.py :: generate_hmm_data(emission_dist='t', clip_factor= ...)
```

Goal: The state labels are encrypted in the synthetic data via HMM. The model's job is to rediscover the labels.



Mixture-VAE code base

1. **State path.** An `hmmlearn` `CategoricalHMM` with `emissionprob_ = I` emits the hidden path $\{S_t\}$ directly. E.g. Transition matrix `[[0.96, 0.04], [0.04, 0.96]]`, regimes persist ~25 days.
2. **Emissions** are **i.i.d. within a state** per state, at $\pm\mathrm{loc}$.
3. **Draw per state.** For each state, draw exactly $|\{t : S_t = j\}|$ values — vectorized, in chunks.
4. **Clip by rejection.** Samples outside $\mathrm{loc}\pm$ `clip_factor` $\times$ **scale** are rejected and redrawn.

Why clip: Undefined moments in small df for Student-*t*. Yuqi used `clip_factor=10.0`.

GAS-SN Enhancement

- `generate_hmm_data(emission_dist='gassn')` at `https://github.com/slihn/Mixture-VAE`

Global regime - Bull/bear states by the jump model

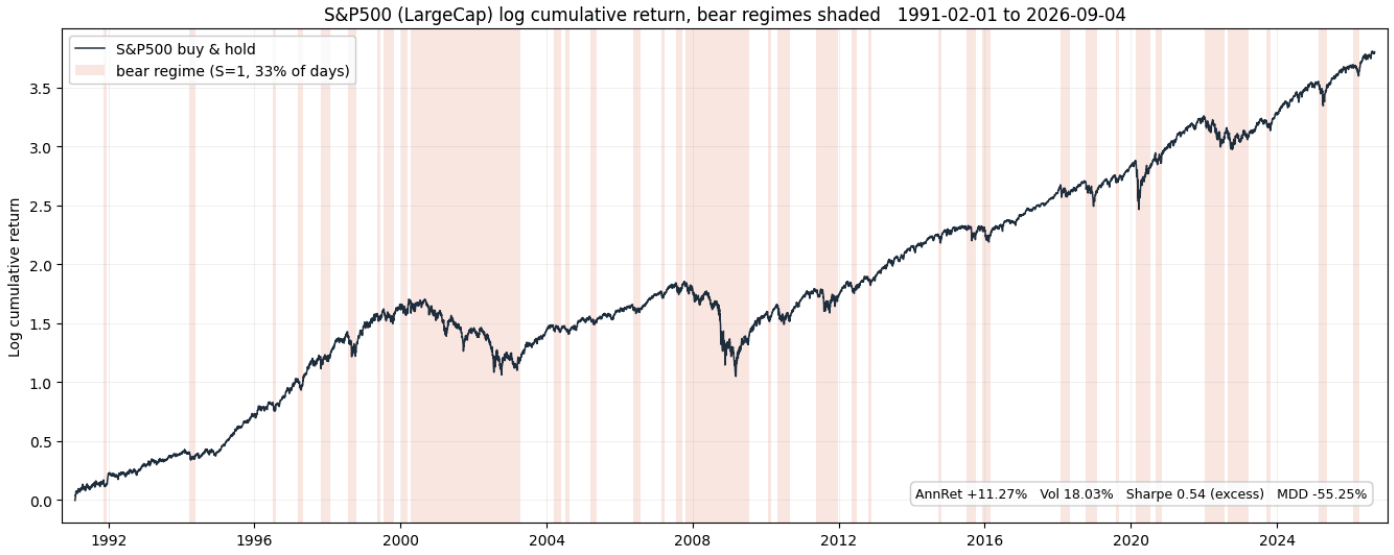
Use the jump model to label the bull/bear states in the S&P500 daily return, $8{,}962$ days, 1991–2026.

Transition Matrix

	bull	bear
bull	0.9940	0.0060
bear	0.0116	0.9884

stationary $(0.659,\ 0.341)$

35 bear episodes over **35** years



Global regime - GAS-SN fits of bull/bear states

Use MLLK to fit the data sets:

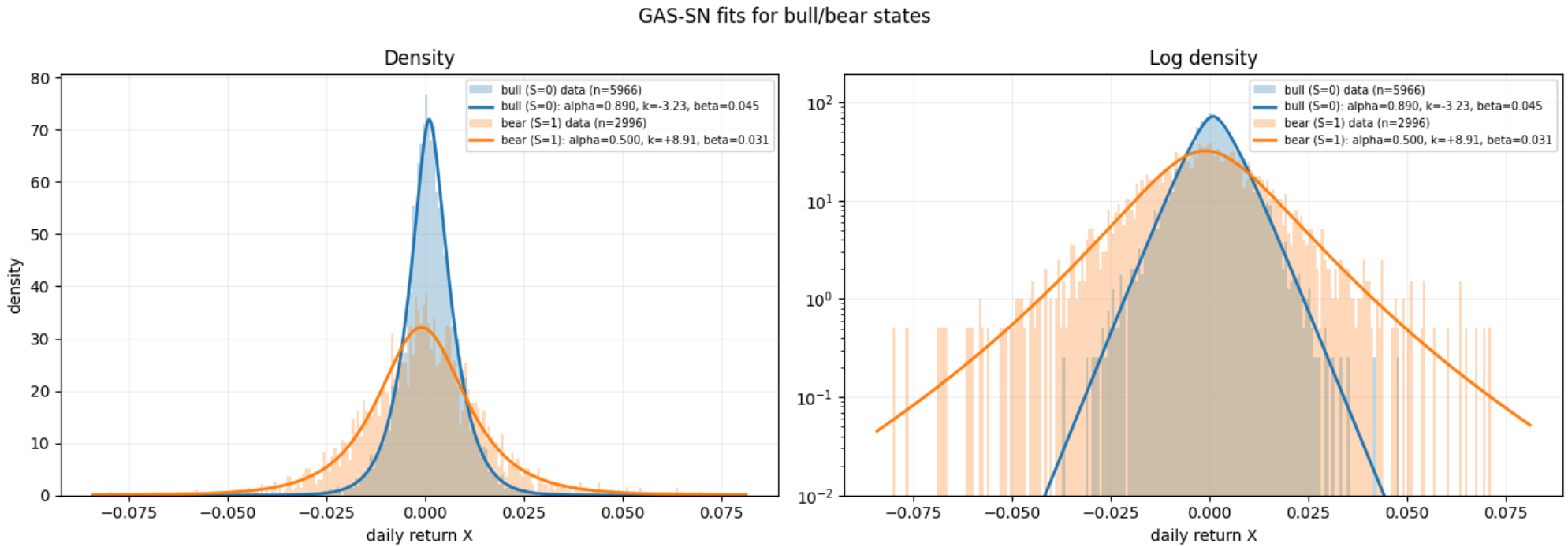
- **bull** in negative-*k*: generalized exponential power (thinner tails)
- **bear** in positive-*k*: generalized α-stable. The large-k and low-alpha region was numerically hard.

state	α	k	β	scale	loc	sd	κ
bull ($S=0$), 66.6%	0.89	$\mathbf{-3.23}$	0.045	0.00536	+0.00089	0.00722	1.86
bear ($S=1$), 33.4%	0.50	8.91	0.031	0.04556	−0.00117	0.01672	5.41

The bear fit was capped at $\alpha = 0.5$ since the kurtosis is fully matched.

The MLE and moment-matching are not aligned.

Global regime - daily return histogram vs theoretical from fits



Simulating from the fits reproduces the real series on share and scale — bear share 0.337 vs 0.334.

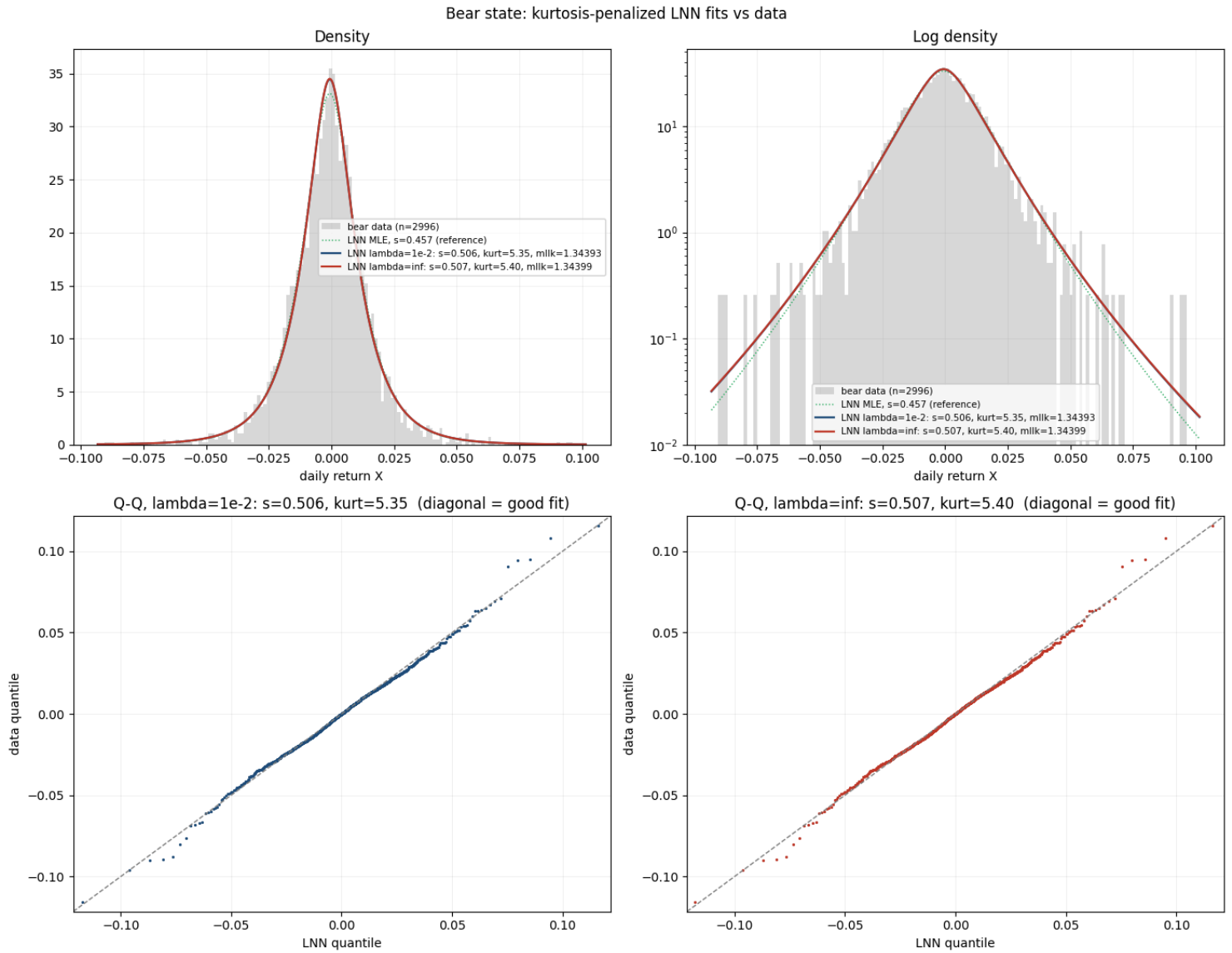
Pooled κ is 8.42 vs data's 10.77, but that gap is **finite-sample noise, not a defect of either fit**: the bear state's 4th moment rides on a handful of extreme draws, and one fit returns κ anywhere from 4.1 to 8.1 across draws.

Global regime - Fit the bear state with LNN

Use three methods to fit $\mathrm{LNN}(\sigma^2)$:

1. Reference is MLE: $\sigma = 0.457$
2. MLE with some penalty to match κ: $\sigma = 0.506$
3. Full penalty to match κ at 5.40: $\sigma = 0.507$

(2) and (3) are shown in Q-Q plots. They are straight lines. These are **good fits**.

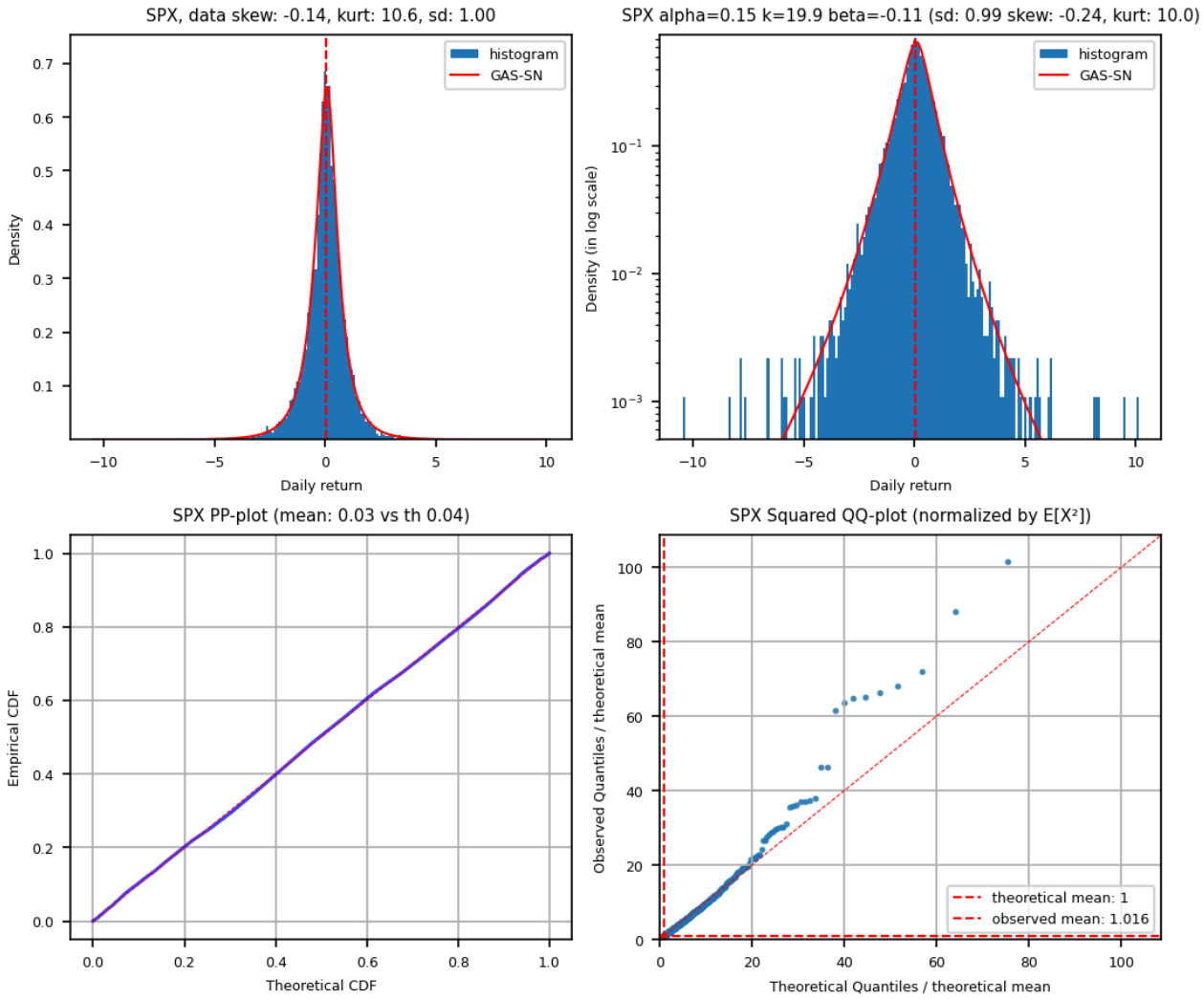


Global regime - New GAS-SN fit of the SPX data (9/2026)

The SPX fit is very good at $\alpha = 0.15, k = 19.9$. Notice the very small α.

This fit was numerically impossible without the new log-inversion algorithm in PDF and CDF.

(SPX daily returns from 1/1990 to mid-2025)



Cluster distance — separation between the two states

$$\text{cluster distance} = \frac{|\text{median}_1 - \text{median}_0|}{\text{MAD}},$$

$$\text{where} \quad \text{MAD} = 1.4826\,\text{median}\big(|x - \text{median}(x)|\big).$$

A better measure of the separation between the two states. It is like the loc/sd, but built from medians — it always exists, and **fat tails cannot move it**.

Rationale: Inside Yuqi's code, X is z-scored before building a feature, so **only the ratio reaches the model**.

Global regime - model output

Cluster distance is only 0.254 MAD — yet Jump scores 0.92.

Feeding those emissions and that transition matrix through the same comparator, $T = 100{,}008$:

model	Accuracy
Jump	**0.9246**
Mixture-VAE	0.8613
KMeans++	0.8269
Gaussian-HMM	0.5000

- All three ML models score very high.
- **HMM fails outright** (0.5000, "model is not converging"), rather than merely trailing.
- Jump ran at `jump_penalty` $= 100$ to match the original notebook.
- Emissions clipped at ± 20 **sd**.

Summary on global regime

- Used the jump model to label the bull/bear states since 1991. 35 bear episodes.
- Fit each state with GAS-SN. Two versions are provided.
- The bear state appears to approach an LNN limit.
- Used GAS-SN as the emission distribution to generate synthetic data.
- Ran four models (Jump, KMeans++, Mixture-VAE, HMM) to label the synthetic data.
- ML model accuracy is high. Jump stands out at 0.92. HMM fails at 0.50.
- A smaller cluster distance is not an obstacle.

These states separate by scale, in addition to opposite locations.

References

- **fracdist** — *Fractional Distributions*, Ch. 1.2 (elements & Table 1), Ch. 6 (FG and the inverse tilted stable law), Ch. 7 (FCM), Ch. 12 (GAS-SN). `github.com/slihn/gas-impl/tree/main/docs`
- Nie, Mulvey, Poor, Yu & Huang (2026) — *Deep generative models meet statistical methods: a generalized framework for financial regime identification*, **Annals of Operations Research** (in press). Code: `github.com/yuqinie98/Mixture-VAE` — the model and benchmark this work builds on.
- Pitman & Yor — Poisson–Dirichlet, the two-parameter CRP, and the α-diversity limit.
- Devroye / Kanter / Zolotarev — the stable-law sampling representation behind Lemma 6.2.
- Koponen (1995), Carr & Geman & Madan & Yor, Rosinski, Massing - Tempered stable law (CGMY)
- Azzalini (2013) — the skew-normal and skew-*t* blueprint (selective sampling, SN_d).